A Long-Awaited Desideratum: MEI Encoding of Byzantine Neumes (Chrysanthine Notation)

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[2]Aristotle University of Thessaloniki
[3]Greek RISM office
[4]McGill University

International Association of Music Libraries (IAML) Congress
30 June 2026

1400 years since the first written testimonium about the Akathistos Hymn

Video by the Christian Student Action of Thessaloniki

https://www.youtube.com/watch?app=desktop&v=4_w6EWdtGHk

Προοίμιον I

Το αρχικό προοίμιο -> Τὸ προσταχθέν

Εφύμνιο I -> Χαῖρε νύμφη ἀνύμφευτε

Προοίμιον II

Το β΄ προοίμιο, νικητήριος ύμνος -> Τῇ Ὑπερμάχῳ

Εφύμνιο I -> Χαῖρε νύμφη ἀνύμφευτε

(Προοίμιον III)

Μεταγενέστερη προσθήκη -> Οὐ παυόμεθα

Εφύμνιο II -> Χαῖρε ἡ Κεχαριτωμένη

ΠΡΟΟΙΜΙΑ

Οίκος Α΄

Ἄγγελος πρωτοστάτης -> Μετρικό και μουσικό μοντέλο για τους υπόλοιπους οίκους με περιττό αριθμό

Χαιρετισμοί (12 στίχοι)

Χαῖρε νύμφη ἀνύμφευτε -> Εφύμνιο I, θεομητορικό

Οίκος Β΄

Βλέπουσα ἡ Ἁγία -> Μετρικό και μουσικό μοντέλο για τους οίκους με άρτιο αριθμό

Ἀλληλούϊα -> Εφύμνιο III, δεσποτικό = αἰνεῖτε τὸν Κύριον

Οίκος Γ΄

Γνῶσιν ἄγνωστον

Χαιρετισμοί (12 στίχοι)

Χαῖρε νύμφη ἀνύμφευτε

Οίκος Δ΄

Δύναμις τοῦ Ὑψίστου

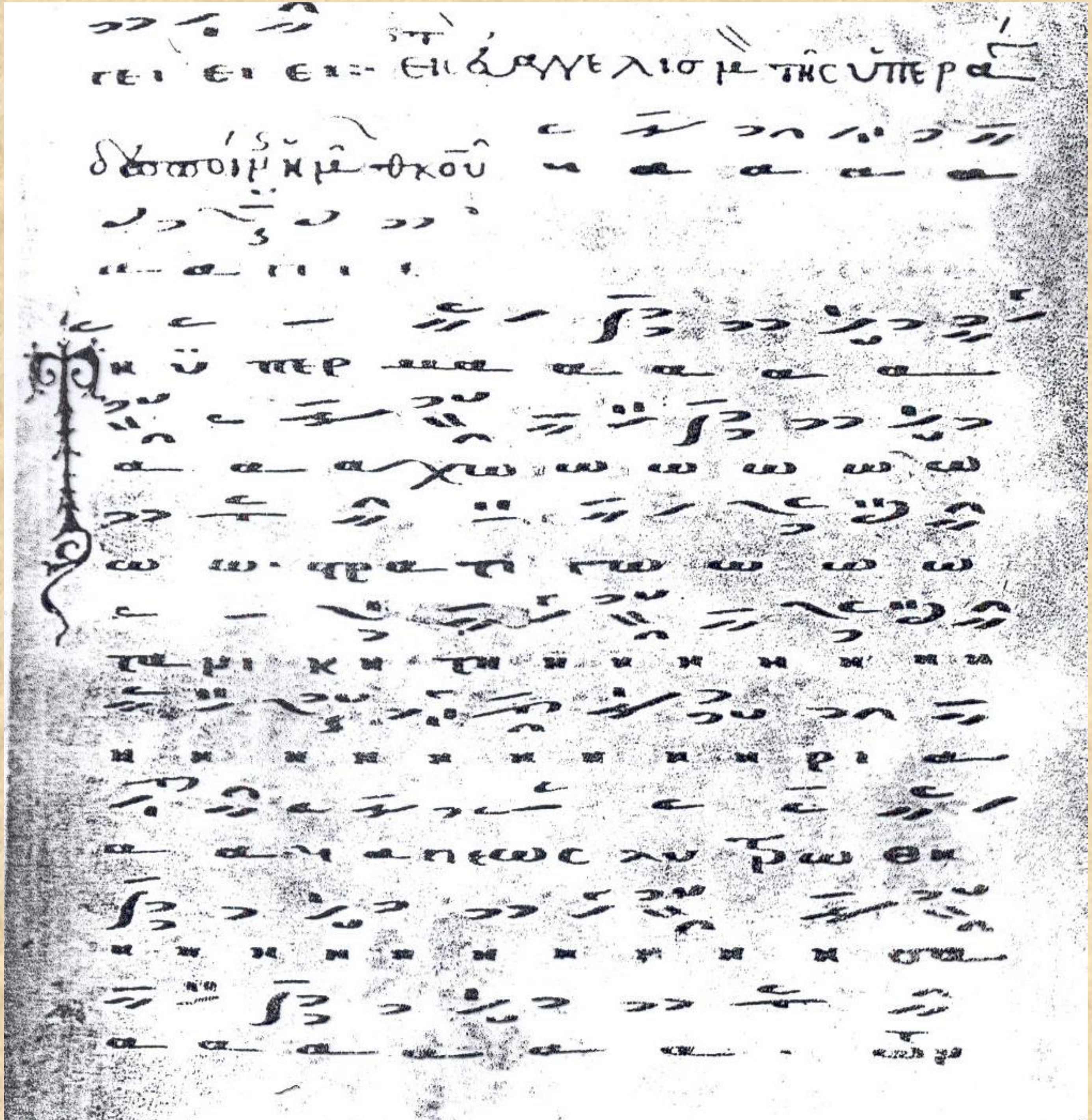
Ἀλληλούϊα

… παρομοίως μέχρι τον Οίκο ΚΔ΄

24 ΟΙΚΟΙ με αλφαβητική ακροστιχίδα ->
Α-Μ: ιστορικό μέρος
Ν-Ω: θεολογικό-δογματικό μέρος

Αρχή της ανώνυμης μελισματικής μελοποίησης ολόκληρου του Ακαθίστου, από το χειρόγραφο Ashburnhamensis L 64, φ. 108α, έτ. 1289, προέλευση: Κρυπτοφέρρη.

Εικόνα από: *Contacarium Ashburnhamense,* επιμ. Carsten Høeg, MMB, Série principale 4 (Copenhague: Munksgaard, 1956).



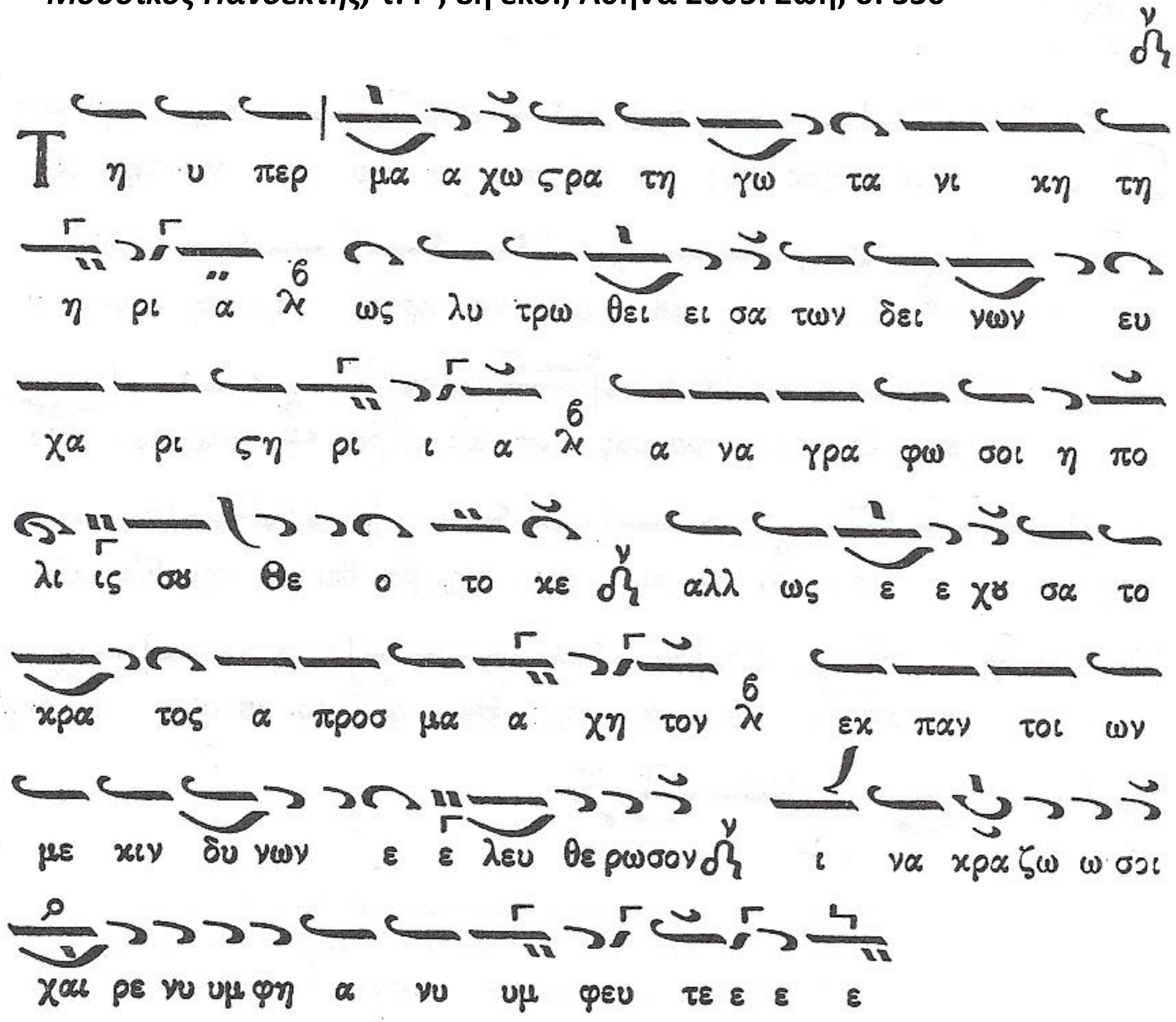
Spyrakou, “When Iconography and *Typika* Meet: Performing the Ἀκάθιστος Ὕμνος in Time and Space” (2025), σσ. 57-78

Musical performance:

1. **Litanies outside the churches,** in the cities usually with the icon of Theotokos Hodegetria (Constantinople, Thessaloniki, Arta), with the participation of chanters, clerics, asketria (women chanters), large parts of the population of all ages, as well as children, singing *Τῇ Ὑπερμάχῳ* κ.ά. (testimonia since the 11th cent.)

Fig. 4.1 Oikoi 23 and 24 in the sanctuary of Marko Monastery; photograph: Elizabeta Dimitrova (with the kind permission of the Ministry of Culture of the Republic of Macedonia, Cultural Heritage Protection Office)

Το β΄ προοίμιον του Ακαθίστου Ύμνου, κείμενο 7ου αι., μέλος ανώνυμο. Πηγή:
***Μουσικός Πανδέκτης,* τ. Γ΄, 8η έκδ., Αθήνα 2005: Ζωή, σ. 336**



A Long-Awaited Desideratum: MEI Encoding of Byzantine Neumes (Chrysanthine Notation)

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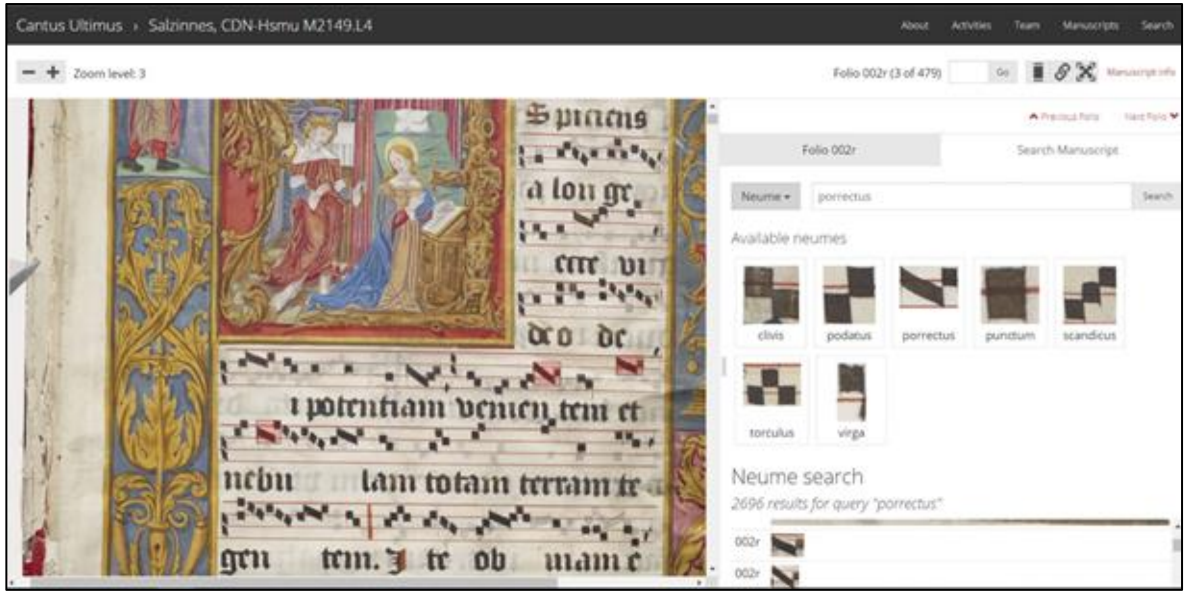
MEI Encoding of Byzantine Neumes
(Chrysanthine Notation)

What is MEI?
And why do we want to use it?

Music Encoding Initiative (MEI)

- It is both a *community* and a *symbolic music format*
- Goal: **define the best practices** to encode a **wide variety** of **music documents**
- **Community:** music librarians and archivists, music theorists, musicologists, and technicians
- **Symbolic format supports the encoding of**:
  - **CWMN** (Common Western Music Notation)
  - **(Western) Early Music**: Western neumes, mensural notation, tablature
  - It is **easily customizable and extensible** (by design) → easy extension to support *other repertoires* and *notations*
  - **Digital editions**
  - **Connections** between the (encoded) music text and the **facsimile**
  - and much more



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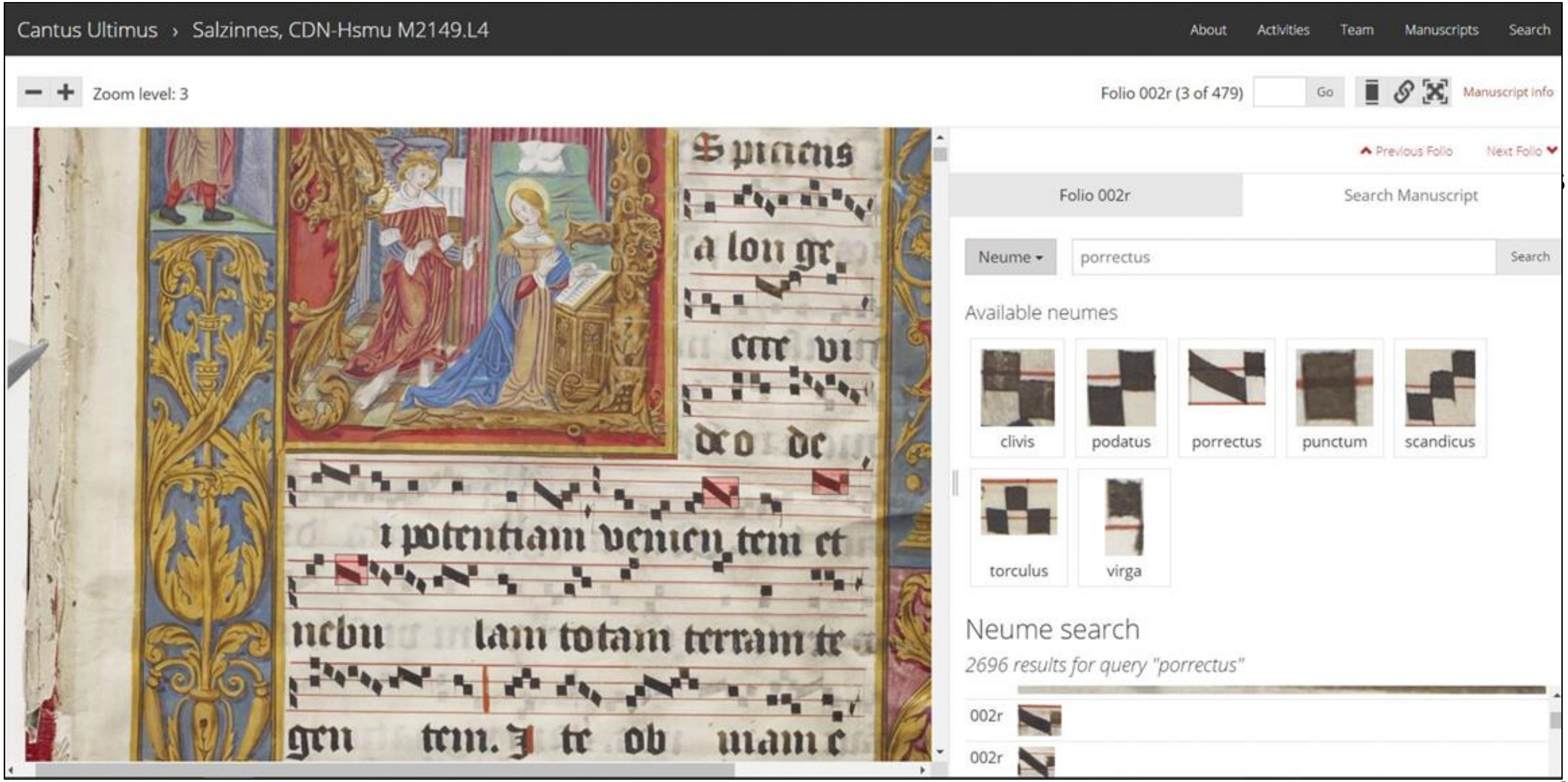
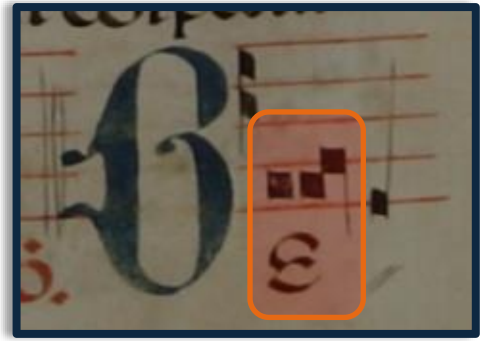


Usual Structure of MEI Neumes Encoding (Western chant)



```
<syllable>
  <syl>Be</syl>
  <neume>
    <nc pname="e" oct="3"/>
    <nc pname="e" oct="3"/>
    <nc pname="f" oct="3" tilt="s"/>
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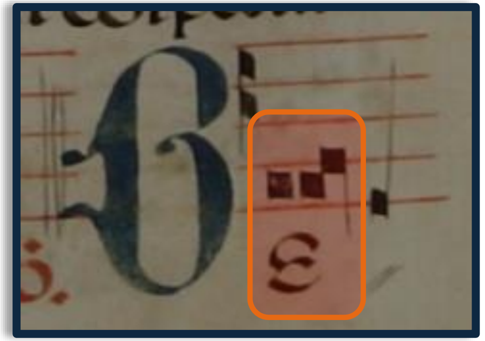
`<syllable>` — **Container element** for everything that belongs to a syllable (neumes and text)



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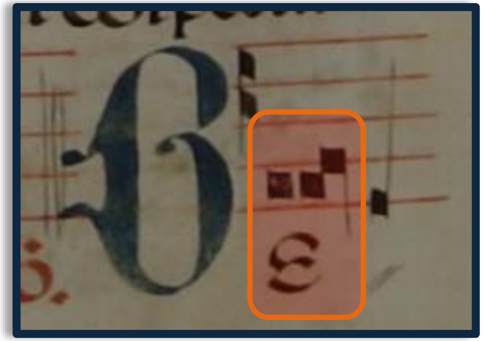

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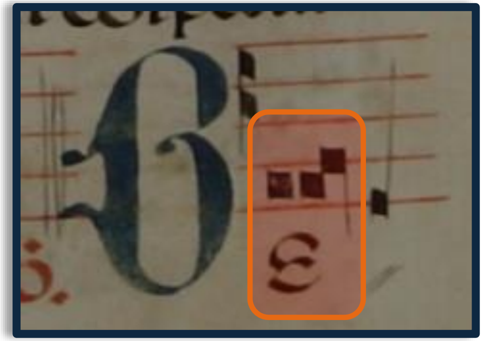

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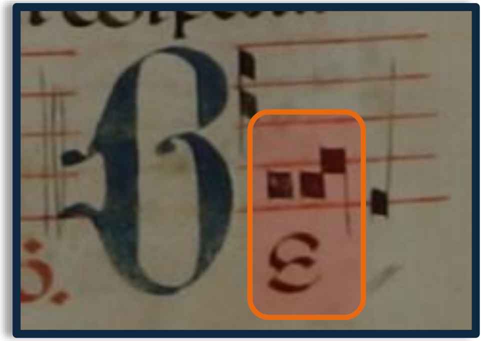

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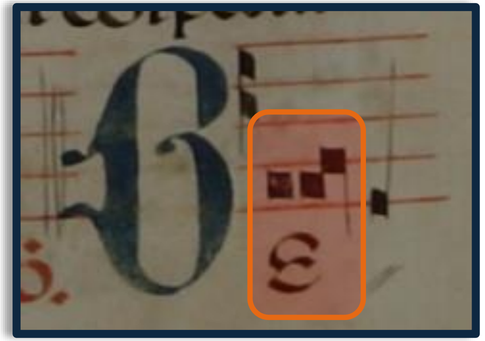
A *neume component* is defined as "sign representing a single pitched event"



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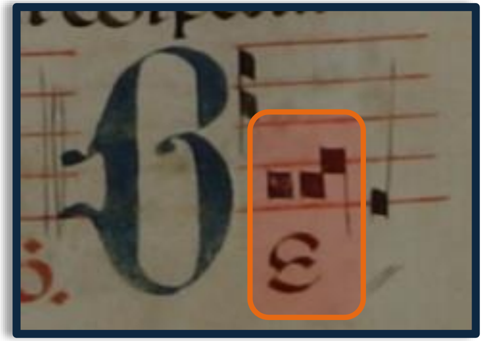
↳ OR **@intm** (*melodic interval* with respect to *previous note*)



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  <syl>Ti</syl>
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Proposal
for the **MEI Encoding**
of **Byzantine Neumes**
of the ***New System***
(in use since 1814)

MEI Encoding for **Byzantine Neumes**

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In **neume combinations**, these attributes appear in the so-called "**parent neume**"

&

their values are derived from the "**basic neumes**" inside the "**parent neume**"

MEI Encoding for **Byzantine Neumes**

combinations
neumes

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combinations neumes


```xml
<syllable>
  <syl>cho</syl>
  <neume dur="2" intmb="-1" intm="-2">
    <neume func="melodic"
     name="apostrofos" pos="main"/>
    <neume func="temporal"
     name="klasma" pos="above"/>
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</syllable>
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- ↳ **Position** (@pos for "basic neumes")

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```

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The “parent neume” contains the pitch and duration information
derived from the “basic (children) neumes” contained within

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TWO EXCEPTIONS

- **Dyo kendimata with oligon**: In this combination of *dyo kendimata above/below oligon*, each individual neume (the *dyo kendimata* and the *oligon*) has its own pitch (and duration)
- **Iporroi**: This "basic neume" represents two notes (two descending voices in scalar movement [-1 -1]). Therefore, each note within the `<neume name="iporroi">` has its own pitch (and duration)

Basic Interval Neumes @func melodic

Shape	Name	Encoding	Description	Interval Sign
	Ison (Íson)	<neume func="melodic" **name="ison" intm="0" intmb="0"**/>	Keep the same step (0)	basic interval sign
	Olígon	<neume func="melodic" **name="oligon" intm="+2" intmb="+1"**/>	One ascending step (+1)	basic interval sign
	Oxeía (Oxía)	<neume func="melodic" **name="oxia" intm="+2" intmb="+1"**/>	One ascending step (+1), with dynamic or ornamental accent	basic interval sign
	Petastḗ (Petastí)	<neume func="melodic" **name="petasti" intm="+2" intmb="+1"**/>	One ascending step (+1), with ornament	basic interval sign
	Dýo kentḗmata (Dío kendímata)	<neume func="melodic" **name="dyo-kendimata" intm="+2" intmb="+1"**/>	One ascending step (+1), with legato	basic interval sign
	Kéntēma (Kéndima)	<neume func="melodic" **name="kendima"**/>	Two ascending steps (+2)	Never stands alone, but in combination with another basic interval sign
	Hypsēlḗ (Ipsilí)	<neume func="melodic" **name="ipsili"**/>	Four ascending steps (+4)	
	Apóstrophos (Apóstrofos)	<neume func="melodic" **name="apostrofos" intm="-2" intmb="-1"**/>	One descending step (-1)	
	Hyporrhoḗ (Iporroí)	<neume func="melodic" **name="iporroi"**> <note **intm="-2" intmb="-1"** slur="m"/> <note **intm="-2" intmb="-1"** slur="t"/> </neume>	Two descending steps in scalar movement (-1-1), with legato starting from the previous note	
	Elaphrón (Elafrón)	<neume func="melodic" **name="elafron" intm="-3" intmb="-2"**/>	Two descending steps leap (-2)	
	Khamēlḗ or Chamēlḗ (Hamilí)	<neume func="melodic" **name="hamili" intm="-5" intmb="-4"**/>	Four descending steps (-4)	

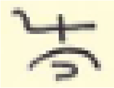
Basic Interval Neumes @func melodic

Shape	Name	Encoding	Description	Interval Sign
	Ison (Íson)	<neume func="melodic" **name="ison" intm="0" intmb="0"**/>	Keep the same step (0)	basic interval sign
	Olígon	<neume func="melodic" **name="oligon" intm="+2" intmb="+1"**/>	One ascending step (+1)	basic interval sign
	Oxeía (Oxía)	<neume func="melodic" **name="oxia" intm="+2" intmb="+1"**/>	One ascending step (+1), with dynamic or ornamental accent	basic interval sign
	Petastḗ (Petastí)	<neume func="melodic" **name="petasti" intm="+2" intmb="+1"**/>	One ascending step (+1), with ornament	basic interval sign
	Dýo kentḗmata (Dío kendímata)	<neume func="melodic" **name="dyo-kendimata" intm="+2" intmb="+1"**/>	One ascending step (+1), with legato	basic interval sign
	Kéntēma (Kéndima)	<neume func="melodic" **name="kendima"**/>	Two ascending steps (+2)	Never stands alone, but in combination with another basic interval sign
	Hypsēlḗ (Ipsilí)	<neume func="melodic" **name="ipsili"**/>	Four ascending steps (+4)	
	Apóstrophos (Apóstrofos)	<neume func="melodic" **name="apostrofos" intm="-2" intmb="-1"**/>	One descending step (-1)	
	Hyporrhoḗ (Iporroí)	<neume func="melodic" **name="iporroi"**> <note **intm="-2" intmb="-1"** slur="m"/> <note **intm="-2" intmb="-1"** slur="t"/> </neume>	Two descending steps in scalar movement (-1-1), with legato starting from the previous note	
	Elaphrón (Elafrón)	<neume func="melodic" **name="elafron" intm="-3" intmb="-2"**/>	Two descending steps leap (-2)	
	Khamēlḗ or Chamēlḗ (Hamilí)	<neume func="melodic" **name="hamili" intm="-5" intmb="-4"**/>	Four descending steps (-4)	

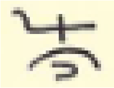
Basic Interval Neumes
@func melodic

Shape	Name	Encoding	Description	Interval Sign
	Ison (Íson)	<neume func="melodic" **name="ison" intm="0" intmb="0"**/>	Keep the same step (0)	basic interval sign
	Olígon	<neume func="melodic" **name="oligon" intm="+2" intmb="+1"**/>	One ascending step (+1)	basic interval sign
	Oxeía (Oxía)	<neume func="melodic" **name="oxia" intm="+2" intmb="+1"**/>	One ascending step (+1), with dynamic or ornamental accent	basic interval sign
	Petastḗ (Petastí)	<neume func="melodic" **name="petasti" intm="+2" intmb="+1"**/>	One ascending step (+1), with ornament	basic interval sign
	Dýo kentḗmata (Dío kendímata)	<neume func="melodic" **name="dyo-kendimata" intm="+2" intmb="+1"**/>	One ascending step (+1), with legato	basic interval sign
	Kéntēma (Kéndima)	<neume func="melodic" **name="kendima"**/>	Two ascending steps (+2)	Never stands alone, but in combination with another basic interval sign
	Hypsēlḗ (Ipsilí)	<neume func="melodic" **name="ipsili"**/>	Four ascending steps (+4)	
	Apóstrophos (Apóstrofos)	<neume func="melodic" **name="apostrofos" intm="-2" intmb="-1"**/>	One descending step (-1)	
	Hyporrhoḗ (Iporroí)	<neume func="melodic" **name="iporroi"**> <note **intm="-2" intmb="-1"** slur="m"/> <note **intm="-2" intmb="-1"** slur="t"/> </neume>	Two descending steps in scalar movement (-1-1), with legato starting from the previous note	
	Elaphrón (Elafrón)	<neume func="melodic" **name="elafron" intm="-3" intmb="-2"**/>	Two descending steps leap (-2)	
	Khamēlḗ or Chamēlḗ (Hamilí)	<neume func="melodic" **name="hamili" intm="-5" intmb="-4"**/>	Four descending steps (-4)	

Compound Interval Neumes **EXAMPLES**

Shape	Name	Meaning	Encoding
	Oligon (Olígon) + Kéntēma (Kéndima) above + Hypsēlḗ (Ipsilí) front right	Six ascending steps of the scale (+6) [ascending seventh]	<neume intm="+7" intmb="+6"> <neume func="melodic" **name="oligon" pos="main"**/> <neume func="melodic" **name="kendima" pos="above"**/> <neume func="melodic" **name="ipsili" pos="front right"**/> </neume>
	Oligon (Olígon) + Kéntēma (Kéndima) above + Hypsēlḗ (Ipsilí) on top	Seven ascending steps of the scale (+7) [ascending octave]	<neume intm="+8" intmb="+7"> <neume func="melodic" **name="oligon" pos="main"**/> <neume func="melodic" **name="kendima" pos="above"**/> <neume func="melodic" **name="ipsili" pos="top"**/> </neume>
	Khamēlḗ or Chamēlḗ (Hamilí) + Elafrón (Elafrón) below + Apóstrophos (Apóstrofos) on the bottom	Seven descending steps of the scale (-7) [descending octave]	<neume intm="-8" intmb="-7"> <neume func="melodic" **name="hamili" pos="main"**/> <neume func="melodic" **name="elafron" pos="below"**/> <neume func="melodic" **name="apostrofos" pos="bottom"**/> </neume>

Compound Interval Neumes **EXAMPLES**

Shape	Name	Meaning	Encoding
	Oligon (Olígon) + Kéntēma (Kéndima) above + Hypsēlḗ (Ipsilí) front right	Six ascending steps of the scale (+6) [ascending seventh]	<neume intm="+7" intmb="+6"> <neume func="melodic" **name="oligon" pos="main"**/> <neume func="melodic" **name="kendima" pos="above"**/> <neume func="melodic" **name="ipsili" pos="front right"**/> </neume>
	Oligon (Olígon) + Kéntēma (Kéndima) above + Hypsēlḗ (Ipsilí) on top	Seven ascending steps of the scale (+7) [ascending octave]	<neume intm="+8" intmb="+7"> <neume func="melodic" **name="oligon" pos="main"**/> <neume func="melodic" **name="kendima" pos="above"**/> <neume func="melodic" **name="ipsili" pos="top"**/> </neume>
	Khamēlḗ or Chamēlḗ (Hamilí) + Elafrón (Elafrón) below + Apóstrophos (Apóstrofos) on the bottom	Seven descending steps of the scale (-7) [descending octave]	<neume intm="-8" intmb="-7"> <neume func="melodic" **name="hamili" pos="main"**/> <neume func="melodic" **name="elafron" pos="below"**/> <neume func="melodic" **name="apostrofos" pos="bottom"**/> </neume>

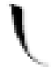
Temporal Neumes

@func temporal

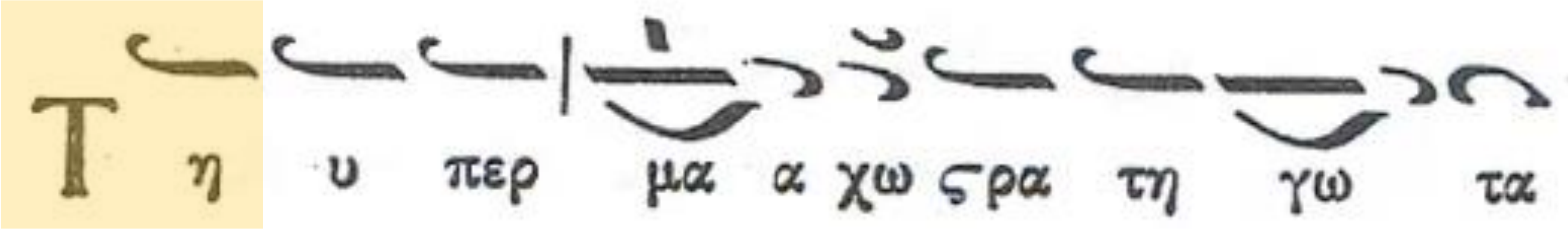
Shape	Name	Encoding	Effect (1 beat = 1 quarter note)
	klasma	<neume func="temporal" **name="klasma"**/>	**Adds** one beat (+1)
	apli	<neume func="temporal" **name="apli"**/>	**Adds** one beat (+1)
	dipli	<neume func="temporal" **name="dipli"**/>	**Adds** two beats (+2)
	tripli	<neume func="temporal" **name="tripli"**/>	**Adds** three beats (+3)
	argon	<neume func="temporal" **name="argon"**/>	**Divides** the first note's beat in two (1/2) **& adds** one beat to the last note (+1)
	triimiargon	<neume func="temporal" **name="triimiargon"**/>	**Divides** the first note's beat in two (1/2) **& adds** two beats to the last note (+2)
	diargon	<neume func="temporal" **name="diargon"**/>	**Divides** the first note's beat in two (1/2) **& adds** three beats to the last note (+3)
	gorgon	<neume func="temporal" **name="gorgon"**/>	**Divides** beat in two (1/2)
	digorgon	<neume func="temporal" **name="digorgon"**/>	**Divides** beat in three (1/3)
	trigorgon	<neume func="temporal" **name="trigorgon"**/>	**Divides** beat in four (1/4)
With Dots			
	dotted digorgon	<neume func="temporal" name="digorgon" dots="1" **dots.gorgon="1" dots.pos="left"**/>	
		<neume func="temporal" name="digorgon" dots="1" **dots.gorgon="2" dots.pos="left"**/>	
		<neume func="temporal" name="digorgon" dots="1" **dots.gorgon="2" dots.pos="right"**/>	

Dynamic/ Ornamental Neumes

@func dynamic/ ornamental

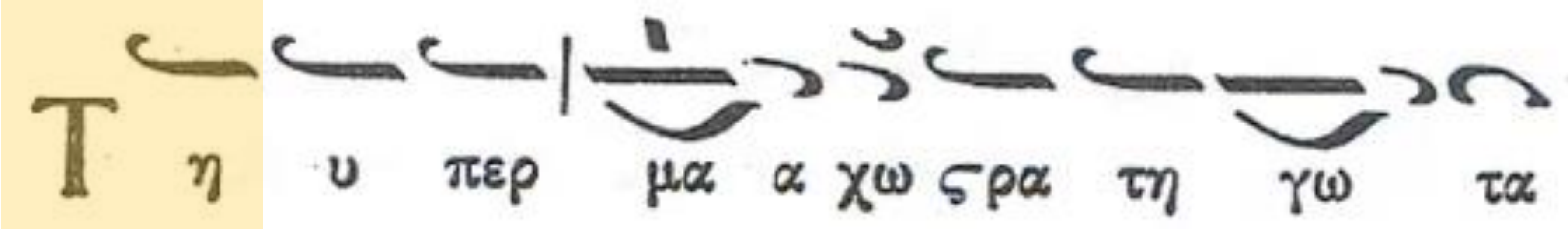
Shape	Name	Encoding	Description
	varia	<neume func="dynamic/ornamental" **name="varia"**/>	with heaviness
	omalon	<neume func="dynamic/ornamental" **name="omalon"**/>	with an ondulation of the voice
	antikenoma	<neume func="dynamic/ornamental" **name="antikenoma"**/>	with a flight / like suspended …
	psifiston	<neume func="dynamic/ornamental" **name="psifiston"**/>	with strength and lively
	eteron	<neume func="dynamic/ornamental" **name="eteron"**/>	legato, with smooth ondulation of the voice
	endofonon	<neume func="dynamic/ornamental" **name="endofonon"**/>	bouche fermée (with a shut mouth)
+	stavros	<neume func="dynamic/ornamental" **name="stavros"**/>	respiration

Example: Ti ypermacho




```xml
<syllable>
   <syl>Ti</syl>
   <neume xml:id="neume001" func="melodic" name="ison"
    dur="4" intmb="0" pname="c"/>
</syllable>
```

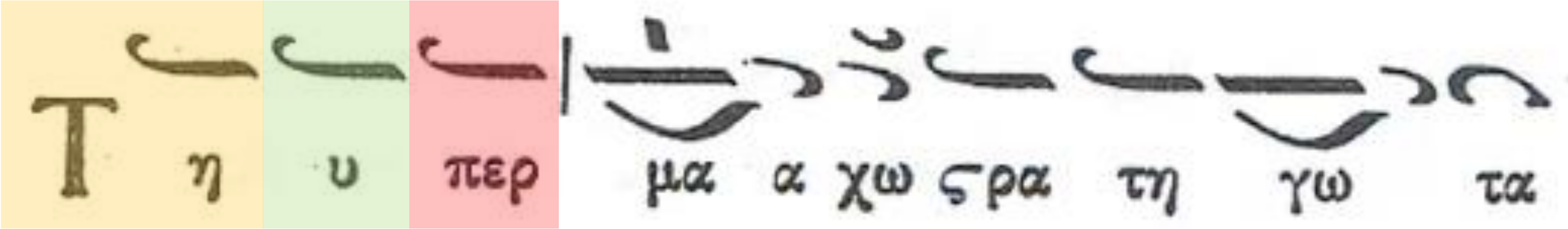
Word:
“Ti”




```xml
<syllable>
   <syl>Ti</syl>
   <neume xml:id="neume001" func="melodic" name="ison"
    dur="4" intmb="0" pname="c"/>
</syllable>
```

Word: “Ti”

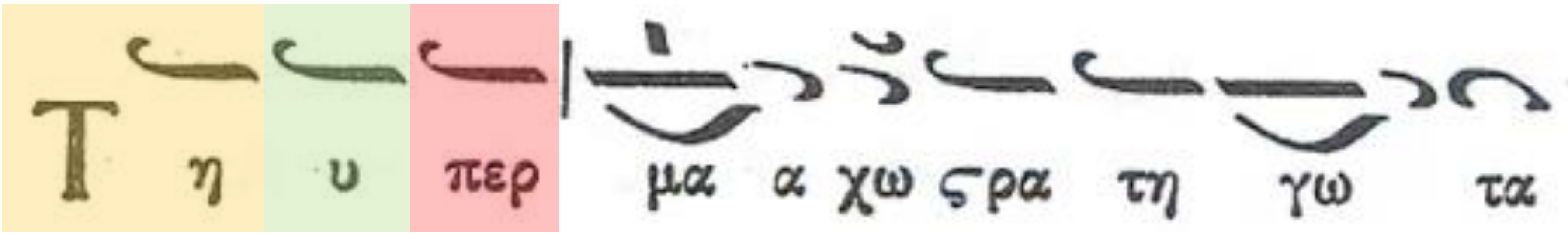
Word: “ypermacho”




```xml
<syllable>
   <syl>Ti</syl>
   <neume xml:id="neume001" func="melodic" name="ison"
    dur="4" intmb="0" pname="c"/>
</syllable>
```

Word: “Ti”

Word: “ypermacho”

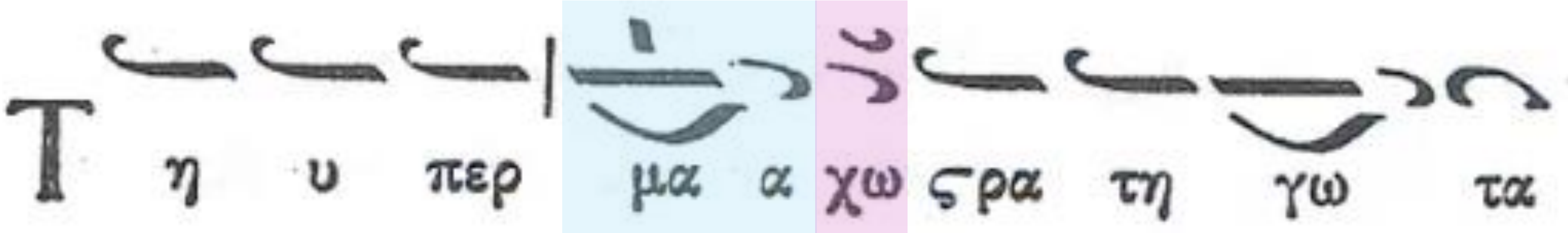



```xml
<syllable>
   <syl>Ti</syl>
   <neume xml:id="neume001" func="melodic" name="ison"
    dur="4" intmb="0" pname="c"/>
</syllable>
```

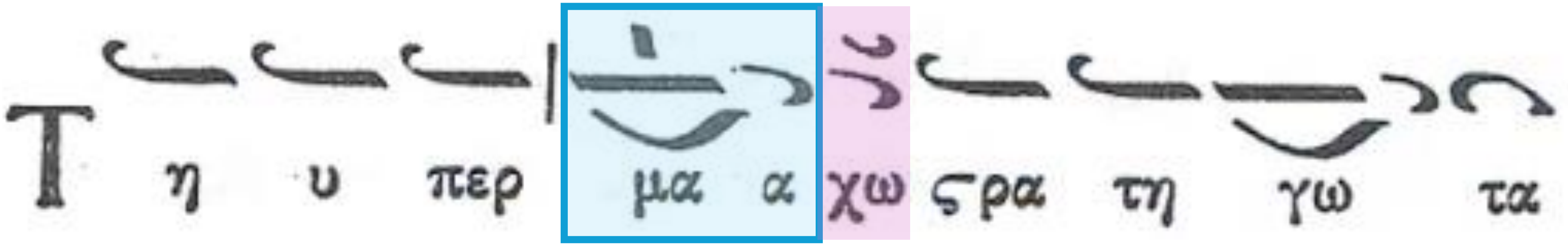
Word: “Ti”

```xml
<syllable>
   <syl>Y-</syl>
   <neume xml:id="neume002" func="melodic" name="ison"
    dur="4" intmb="0" pname="c"/>
</syllable>
<syllable>
   <syl>per-</syl>
   <neume xml:id="neume003" func="melodic" name="ison"
    dur="4" intmb="0" pname="c"/>
</syllable>
```

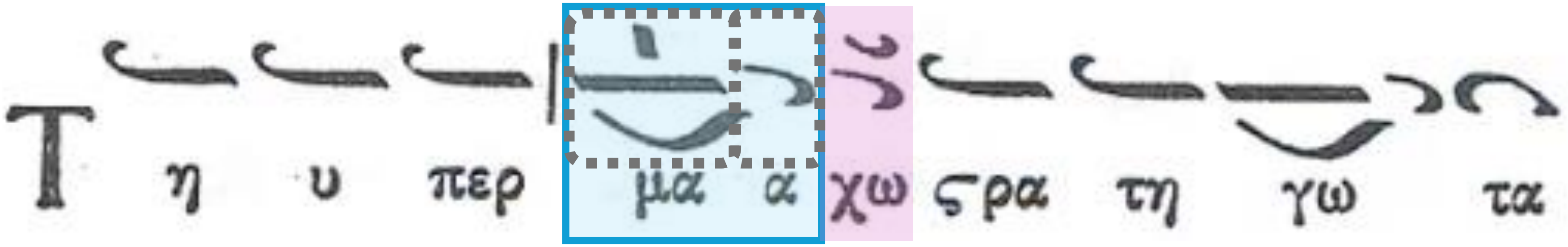
Word: “ypermacho”



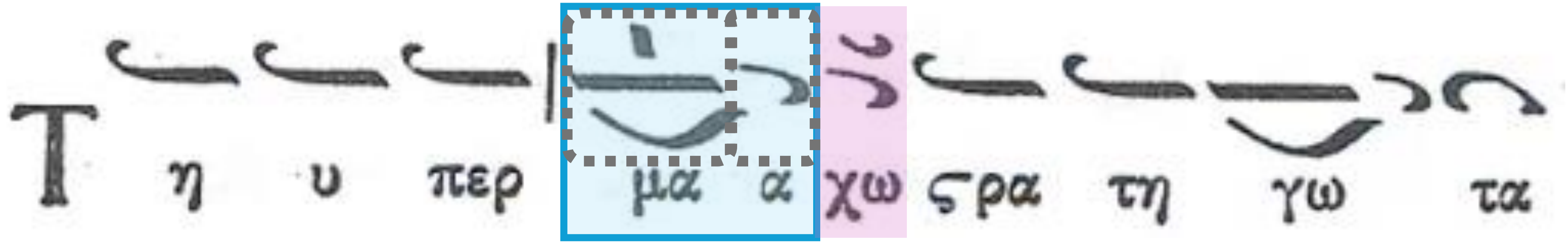
continue with word “ypermacho”



continue with word “ypermacho”

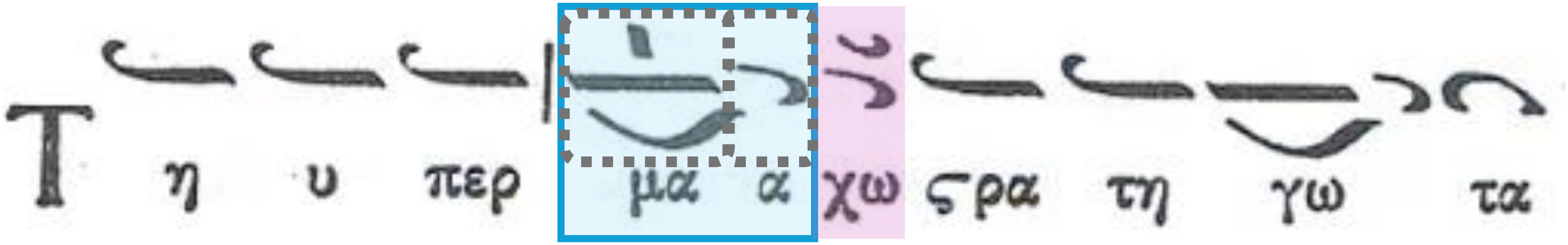


continue with word “ypermacho”



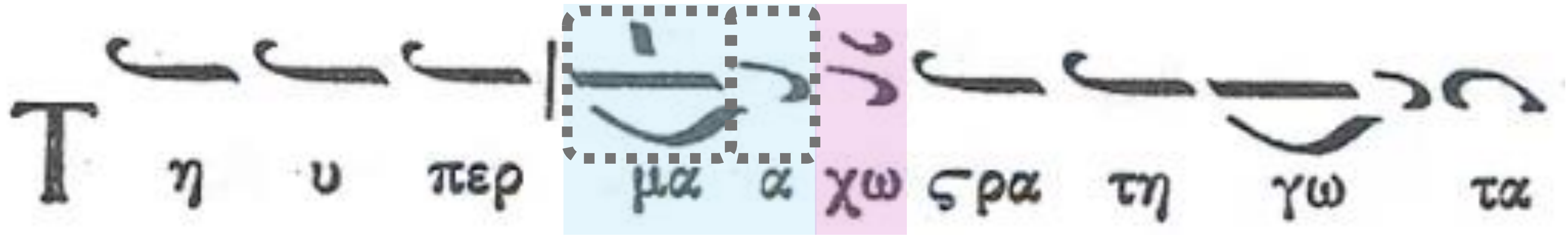
continue with word “ypermacho”

```xml
<syllable>
   <syl>ma-</syl>
   <neume xml:id="neume004" dur="4" intmb="+3" intm="+4" pname="f">
      <neume xml:id="neume005" func="melodic" name="oligon" pos="main"/>
      <neume xml:id="neume006" func="melodic" name="kendima" pos="above"/>
      <neume xml:id="neume007" func="dynamic/ornamental" name="psifiston" pos="below"/>
   </neume>
   <neume xml:id="neume008" func="melodic" name="apostrofos" dur="4" intmb="-1"
    intm="-2" pname="e"/>
</syllable>
```



continue with word “ypermacho”

```
<syllable>
   <syl>ma-</syl>
   <neume xml:id="neume004" dur="4" intmb="+3" intm="+4" pname="f">
      <neume xml:id="neume005" func="melodic" name="oligon" pos="main"/>
      <neume xml:id="neume006" func="melodic" name="kendima" pos="above"/>
      <neume xml:id="neume007" func="dynamic/ornamental" name="psifiston" pos="below"/>
   </neume>
   <neume xml:id="neume008" func="melodic" name="apostrofos" dur="4" intmb="-1"
    intm="-2" pname="e"/>
</syllable>
```

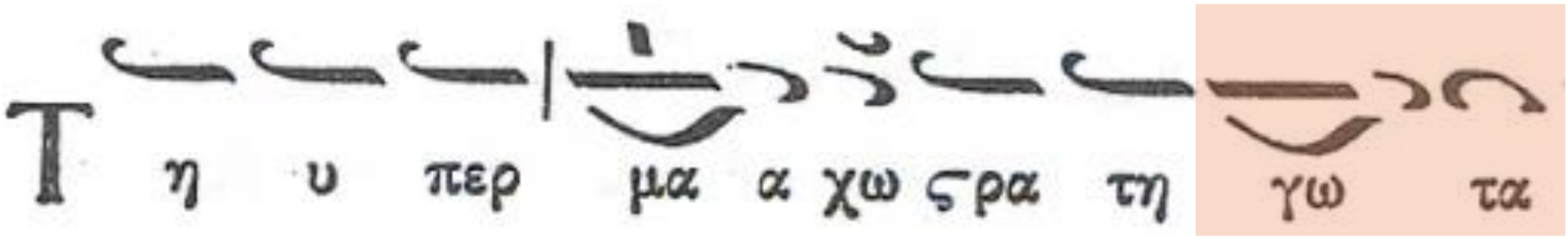


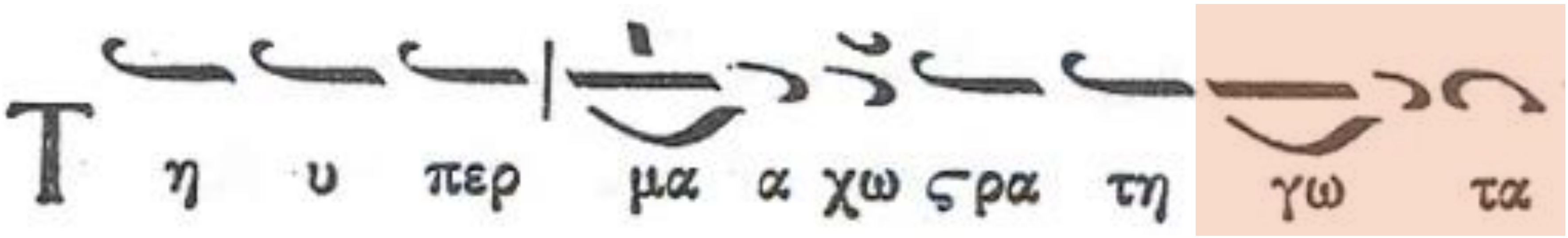
continue with word “ypermacho”

```
<syllable>
   <syl>ma-</syl>
   <neume xml:id="neume004" dur="4" intmb="+3" intm="+4" pname="f">
      <neume xml:id="neume005" func="melodic" name="oligon" pos="main"/>
      <neume xml:id="neume006" func="melodic" name="kendima" pos="above"/>
      <neume xml:id="neume007" func="dynamic/ornamental" name="psifiston" pos="below"/>
   </neume>
   <neume xml:id="neume008" func="melodic" name="apostrofos" dur="4" intmb="-1"
    intm="-2" pname="e"/>
</syllable>
```

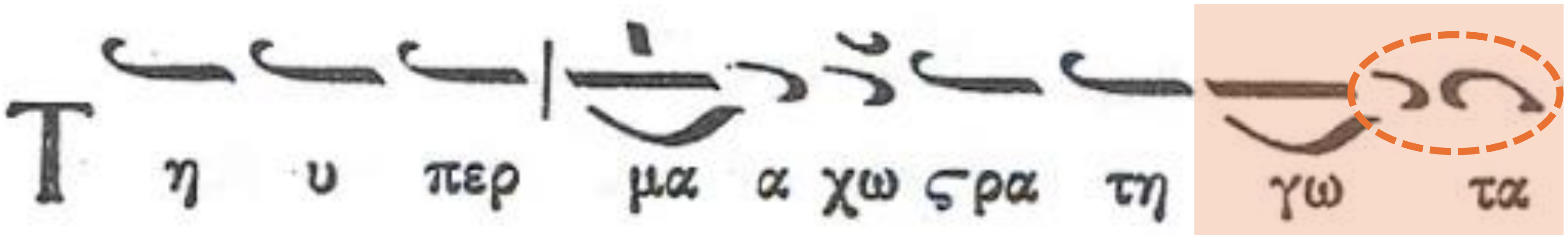
```
<syllable>
   <syl>cho</syl>
   <neume xml:id="neume009" dur="2" intmb="-1" intm="-2" pname="d">
      <neume xml:id="neume010" func="melodic" name="apostrofos" pos="main"/>
      <neume xml:id="neume011" func="temporal" name="klasma" pos="above"/>
   </neume>
</syllable>
```

We have already seen this one

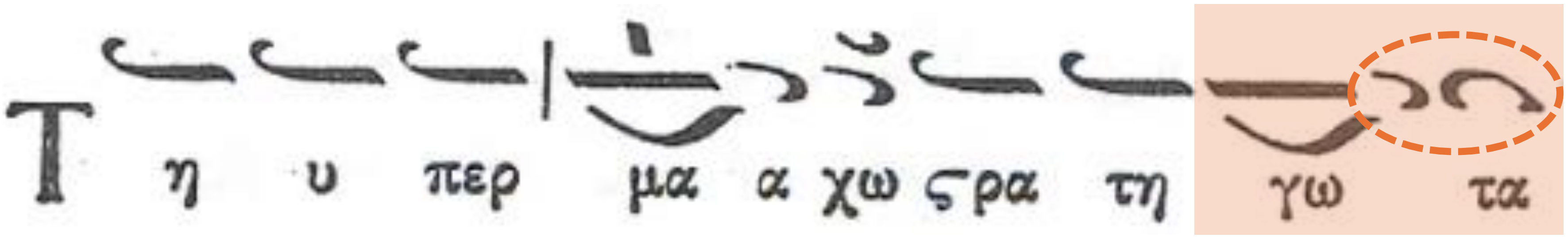

Τ η υ περ μα α χω ϛρα τη γω τα



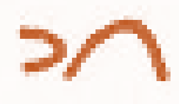
→ formulas



→ formulas

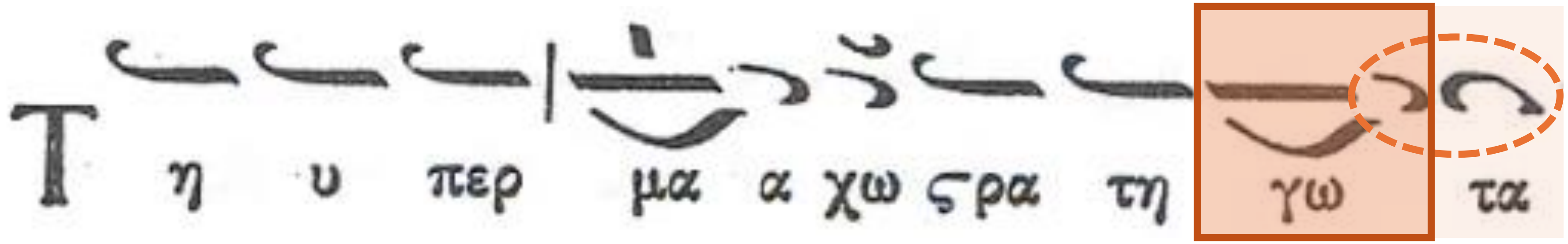


→ formulas

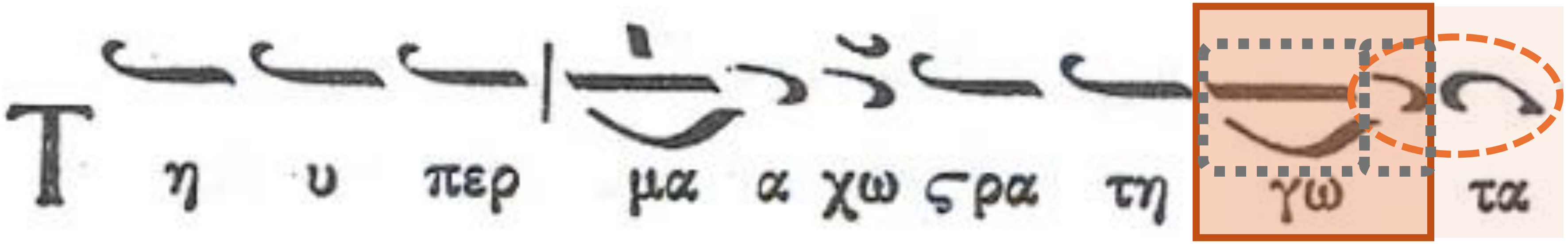
	**Syneches Elafron**

*Two descending steps [-1 -1], so two notes in downward scalar movement, with **the first note (the apostrofos)** having a duration of an **eighth note** and **lying on the same beat (quarter note) as the preceding note***

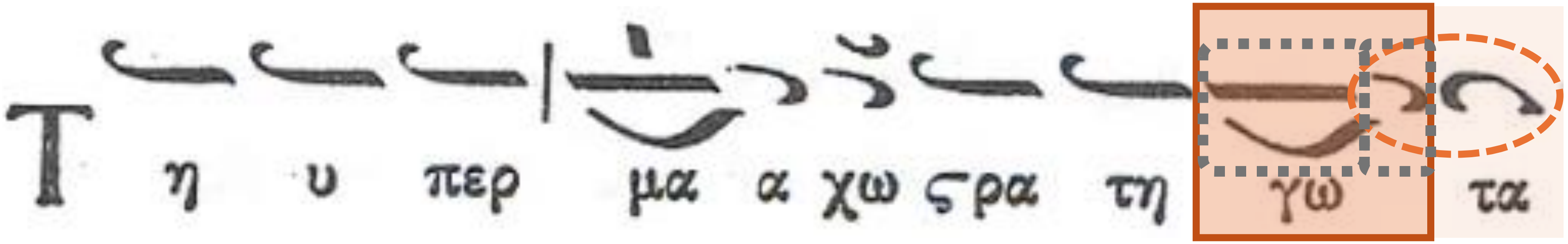
& with the second note of the scalar movement being a quarter note



→ formulas



→ formulas



→ formulas

```
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   <syl>go</syl>
   <neume xml:id="neume014" dur="8" intmb="+1" intm="+2" pname="e" slur="i">
      <neume xml:id="neume015" func="melodic" name="oligon" pos="main"/>
      <neume xml:id="neume016" func="dynamic/ornamental" name="psifiston" pos="below"/>
   </neume>
   <neume xml:id="neume017" func="melodic" name="apostrofos-of-syneches-elafron"
    dur="8" intmb="-1" intm="-2" pname="d" slur="t"/>
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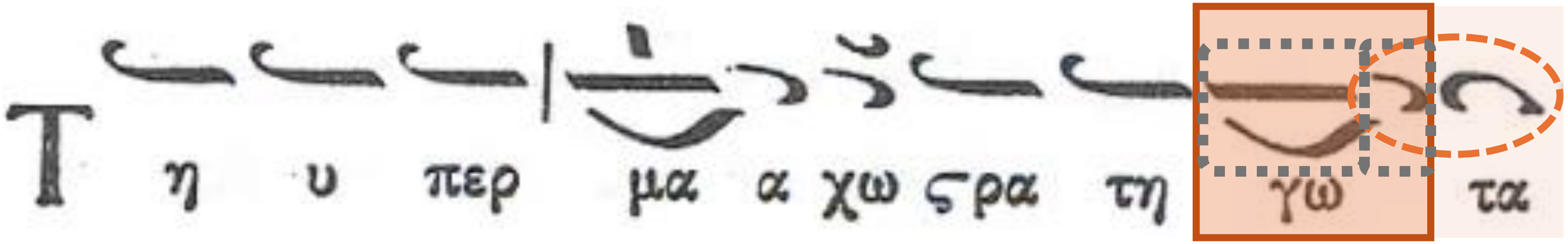
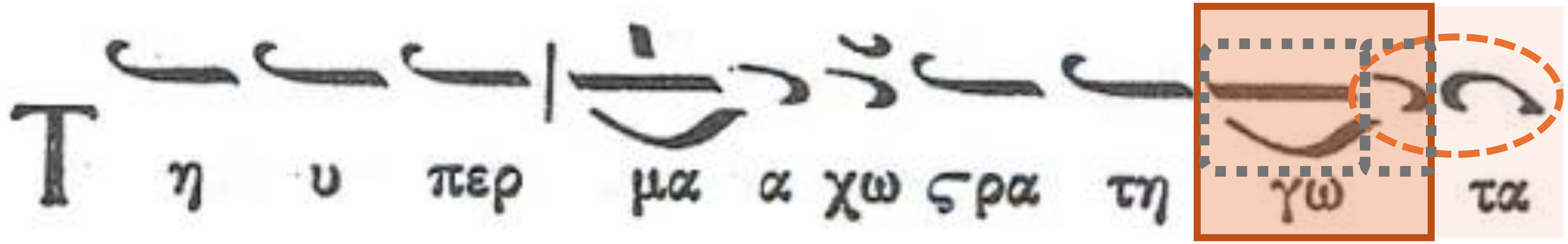


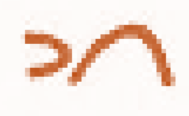
→ formulas

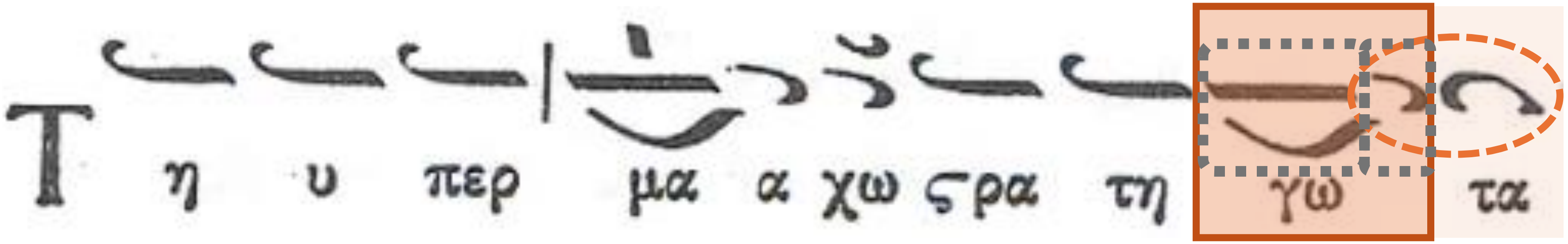
```xml
<syllable>
   <syl>go</syl>
   <neume xml:id="neume014" dur="8" intmb="+1" intm="+2" pname="e" slur="i">
      <neume xml:id="neume015" func="melodic" name="oligon" pos="main"/>
      <neume xml:id="neume016" func="dynamic/ornamental" name="psifiston" pos="below"/>
   </neume>
   <neume xml:id="neume017" func="melodic" name="apostrofos-of-syneches-elafron"
    dur="8" intmb="-1" intm="-2" pname="d" slur="t"/>
</syllable>
```



→ formulas

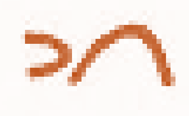
```xml
<syllable>
   <syl>go</syl>
   <neume xml:id="neume014" dur="8" intmb="+1" intm="+2" pname="e" slur="i">
      <neume xml:id="neume015" func="melodic" name="oligon" pos="main"/>
      <neume xml:id="neume016" func="dynamic/ornamental" name="psifiston" pos="below"/>
   </neume>
   <neume xml:id="neume017" func="melodic" name="apostrofos-of-syneches-elafron"
    dur="8" intmb="-1" intm="-2" pname="d" slur="t"/>
</syllable>
```

	**Syneches Elafron**
	*Two descending steps [-1 -1], so two notes in downward scalar movement, with **the first note (the apostrofos)** having a duration of an **eighth note** and **lying on the same beat (quarter note) as the preceding note** & with the second note of the scalar movement being a quarter note*



→ formulas

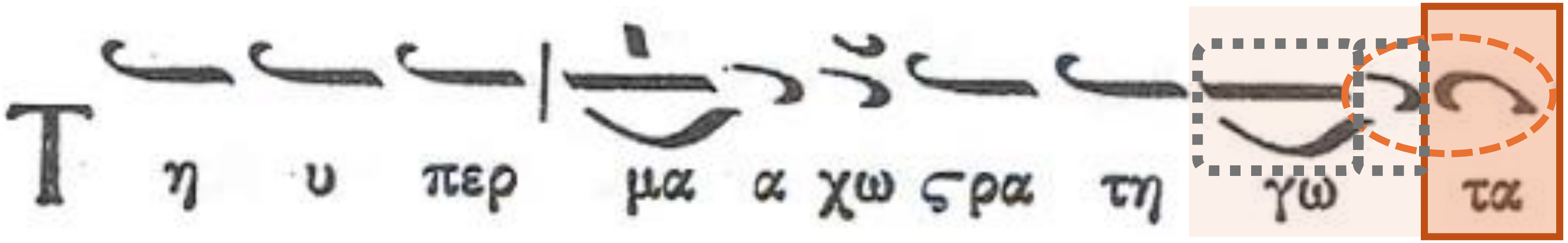
```
<syllable>
   <syl>go</syl>
   <neume xml:id="neume014" dur="8" intmb="+1" intm="+2" pname="e" slur="i">
      <neume xml:id="neume015" func="melodic" name="oligon" pos="main"/>
      <neume xml:id="neume016" func="dynamic/ornamental" name="psifiston" pos="below"/>
   </neume>
   <neume xml:id="neume017" func="melodic" name="apostrofos-of-syneches-elafron"
    dur="8" intmb="-1" intm="-2" pname="d" slur="t"/>
</syllable>
```



Syneches Elafron

Two descending steps [-1 -1], *so two notes in downward scalar movement, with the first note (the apostrofos) having a duration of an eighth note and lying on the same beat (quarter note) as the preceding note*

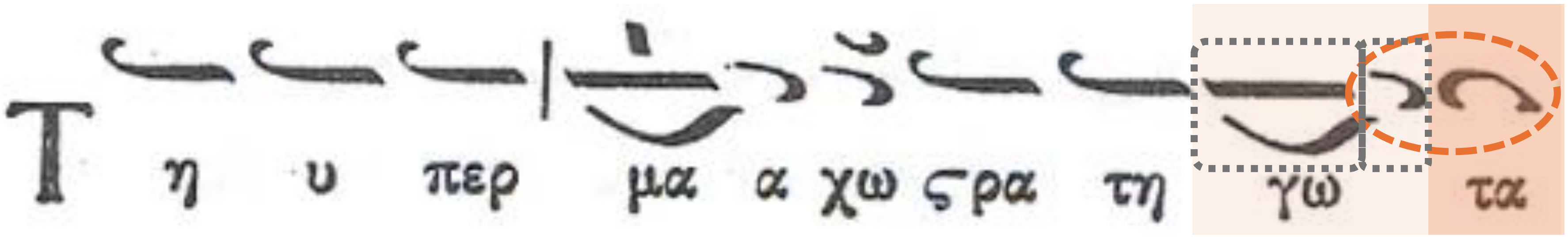
& with the ***second note of the scalar movement being a quarter note***



→ formulas

```xml
<syllable>
   <syl>go</syl>
   <neume xml:id="neume014" dur="8" intmb="+1" intm="+2" pname="e" slur="i">
      <neume xml:id="neume015" func="melodic" name="oligon" pos="main"/>
      <neume xml:id="neume016" func="dynamic/ornamental" name="psifiston" pos="below"/>
   </neume>
   <neume xml:id="neume017" func="melodic" name="apostrofos-of-syneches-elafron"
    dur="8" intmb="-1" intm="-2" pname="d" slur="t"/>
</syllable>
```

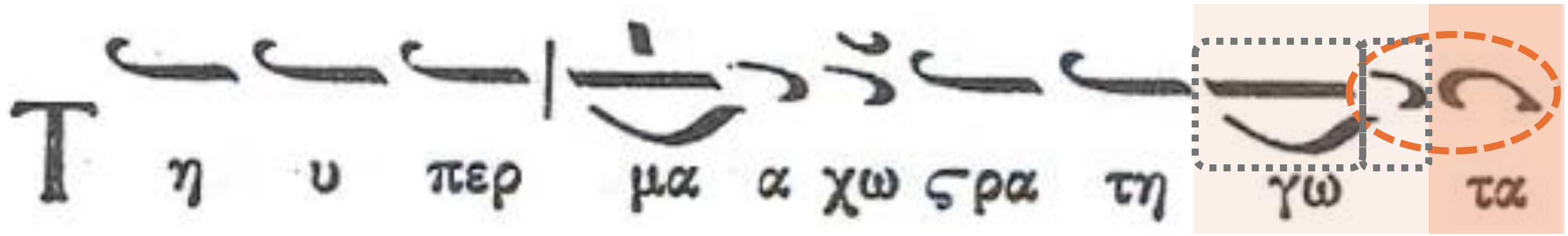
```xml
<syllable>
   <syl>ta</syl>
   <neume xml:id="neume018" func="melodic" name="syneches-elafron" dur="4" intmb="-1"
    intm="-2" pname="c"/>
</syllable>
```



→ formulas

```xml
<syllable>
   <syl>go</syl>
   <neume xml:id="neume014" dur="8" intmb="+1" intm="+2" pname="e" slur="i">
      <neume xml:id="neume015" func="melodic" name="oligon" pos="main"/>
      <neume xml:id="neume016" func="dynamic/ornamental" name="psifiston" pos="below"/>
   </neume>
   <neume xml:id="neume017" func="melodic" name="apostrofos-of-syneches-elafron"
    dur="8" intmb="-1" intm="-2" pname="d" slur="t"/>
</syllable>
```

```xml
<syllable>
   <syl>ta</syl>
   <neume xml:id="neume018" func="melodic" name="syneches-elafron" dur="4" intmb="-1"
    intm="-2" pname="c"/>
</syllable>
```



→ formulas

```xml
<syllable>
   <syl>go</syl>
   <neume xml:id="neume014" dur="8" intmb="+1" intm="+2" pname="e" slur="i">
      <neume xml:id="neume015" func="melodic" name="oligon" pos="main"/>
      <neume xml:id="neume016" func="dynamic/ornamental" name="psifiston" pos="below"/>
   </neume>
   <neume xml:id="neume017" func="melodic" name="apostrofos-of-syneches-elafron"
    dur="8" intmb="-1" intm="-2" pname="d" slur="t"/>
</syllable>
```

```xml
<syllable>
   <syl>ta</syl>
   <neume xml:id="neume018" func="melodic" name="syneches-elafron" dur="4" intmb="-1"
    intm="-2" pname="c"/>
</syllable>
```

```xml
<formula type="syneches-elafron" startid="#neume014" endid="#neume018"/>
```

Encoding of Byzantine Neumes in MEI

We covered the encoding of various types of neumes:

- Melodic, temporal, and dynamic/ornamental
- How to encode “basic neumes”
- How to encode “neume combinations”
- Where to include the duration and pitch information

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Other features that are also supported: mode signatures, tone signatures, attractions, and more formulas

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for the full encoding of the *Ti ypermacho*:
https://github.com/ECHOES-from-the-Past/mei-byzantine-neumes/
(*see handout*)

The Future

- **Automatic transcription into staff notation:** Everything needed is in the MEI Byzantine Neumes file (e.g., pitch and duration information)
- **Optical Music Recognition (OMR):** There is already an encoding system (MEI) for the OMR to generate
- **Computational music analysis:** MEI file is a machine-readable file
  Examples: music density, pitches, recurring patterns
- **Long term:** Digital editions (with the music text linked to facsimile)
- **Long term:** Big data analysis

Thank you ☺

- Echoes from the Past: Unveiling a Lost Soundscape with Digital Analysis (2022.01957.PTDC)
    - Digital Analysis of Chant Transmission - SSHRC Partnership Grant (895-2023-1002)

CENTRO DE ESTUDOS EM MÚSICA CESEM
NOVAFCSH FACULDADE DE CIÊNCIAS SOCIAIS E HUMANAS UNIVERSIDADE NOVA DE LISBOA
IN2PAST PATRIMÓNIO | ARTE | SUSTENTABILIDADE | TERRITÓRIO
fct Fundação para a Ciência e a Tecnologia
ARISTOTLE UNIVERSITY OF THESSALONIKI
DACT
Social Sciences and Humanities Research Council of Canada
Conseil de recherches en sciences humaines du Canada
Canada

Extra Slides (for Q&A)

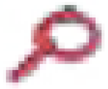
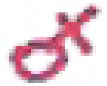
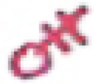
Encoding of other notational symbols:

accidentals, mode signature, key signature

Attractions

- Encoded as **attributes**:
  - `@accid`: written accidental
  - `@accid.ges`: implicit accidental (i.e., implied by a tone signature)
- **To be used in `<neume>`**

  OR in a child `<accid>` when more attributes need to be provided (e.g., `@pos` to indicate the position of the accidental relative to the "main neume")

shape	meaning	encoding
	2 microtones flat (out of twelve microtones)	@accid = 2twf
	4 microtones flat (out of twelve microtones)	@accid = 4twf
	6 microtones flat (out of twelve microtones)	@accid = 6twf
	8 microtones flat (out of twelve microtones)	@accid = 8twf
	2 microtones sharp (out of twelve microtones)	@accid = 2tws
	4 microtones sharp (out of twelve microtones)	@accid = 4tws
	6 microtones sharp (out of twelve microtones)	@accid = 6tws
	8 microtones sharp (out of twelve microtones)	@accid = 8tws

Mode signature: <modeSig>

- Beginning of the piece
- First element in <layer>, preceding all the <syllable> elements

```
<modeSig mode.num="4" mode="plagal" pname="c" fthora="true"
genre="diatonic" nuance="soft"/>
```

- Attributes:
  - @mode.num = 1, 2, 3, 4
  - @mode = authentic, plagal (extend current values)
  - @pname = b (*Zo*), c (*Ni*), d (*Pa*), e (*Vu*), f (*Ga*), g (*Di*), a (*Ke*)
  - @fthora = true, false
  - @genre = diatonic, chromatic, enharmonic
  - @nuance = soft, tense
  - @mode.branch (optional) = diphonos, triphonos, pentaphonos, eptaphonos , antiphonos, naos

Tone signature: <toneSig>

- Child of <layer>
- To be used when the tone signature **changes:**
  - Preceded by the <syllable> elements of the previous tone
  - Followed by the <syllable> elements in the new tone

```
<toneSig pname="e" register="middle"
 genre="diatonic" nuance="soft"/>
```

- Attributes:
  - @pname = b (*Zo*), c (*Ni*), d (*Pa*), e (*Vu*), f (*Ga*), g (*Di*), a (*Ke*)
  - @register = low, middle, high
  - @genre = diatonic, chromatic, enharmonic
  - @nuance = soft, tense

Extra Slides (for Q&A)

The two edge cases: *dyo kendimata with oligon* and *iporroi*

Special case: dyo kendimata with oligon

- **Exception! It is not the “parent neume” that contains the pitch and duration information**
- This is because: in a *dyo kendimata with oligon,* either above or below, each neume (‘oligon’ and ‘dyo kendimata’) has its own pitch and duration

```
<neume xml:id="neume084">
  <neume func="melodic" name="oligon" pos="main"
   dur="4" intm="+2" intmb="+1" pname="d" slur="i"/>
  <neume func="melodic" name="dyo-kendimata" pos="above"
   dur="4" intm="+2" intmb="+1" pname="e" slur="t"/>
</neume>
```

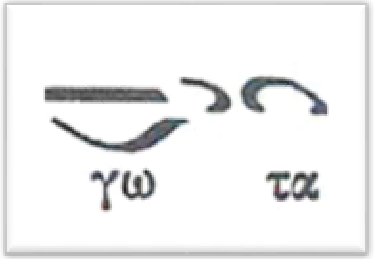
Extra Slides (for Q&A)

Final cadence & other formulas

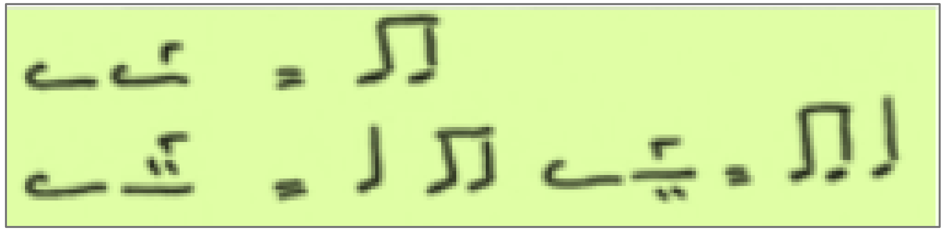
Formulas

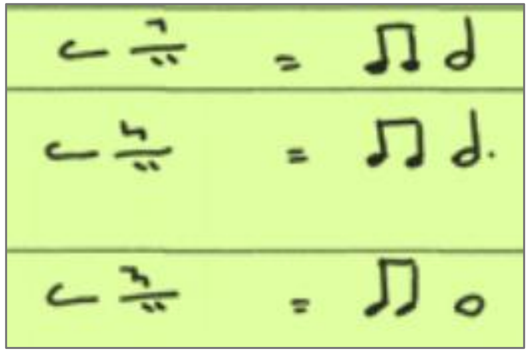
```
<formula startid="#REF1" endid="#REF2"/>
```

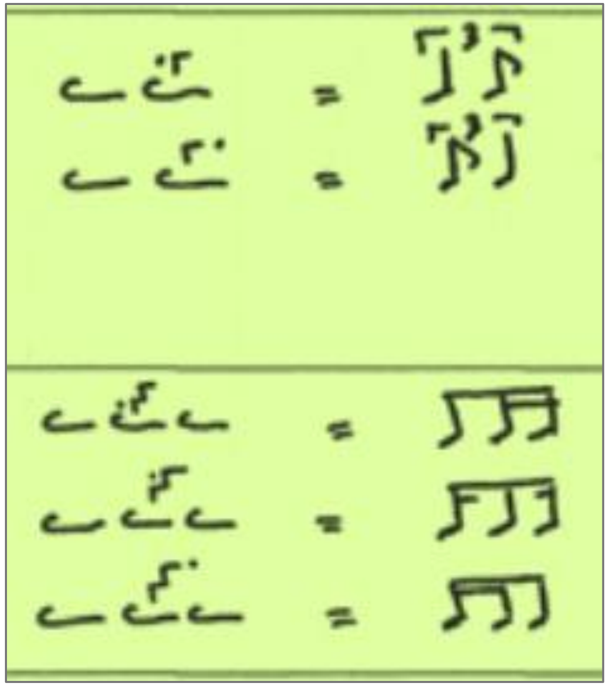
- Syneches elafron:



- With gorgon or argon:

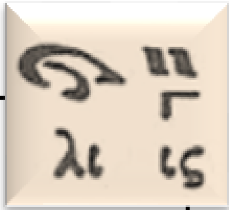






Formulas

```
<formula startid="#REF1" endid="#REF2"/>
```



```
<syllable>
  <syl>lis</syl>
start<neume xml:id="neume072" dur="8" intmb="-3" intm="-4" pname="d" slur="i">
    <neume xml:id="neume073" func="melodic" name="elafron" pos="main"/>
    <neume xml:id="neume074" func="melodic" name="apostrofos" pos="below"/>
  </neume>
end<neume xml:id="neume075" intmb="+1" intm="+2" pname="e" dur="8" slur="t">
    <neume xml:id="neume076" func="melodic" name="dio kendimata" pos="main"/>
    <neume xml:id="neume077" func="temporal" name="gorgon" pos="below"
     effect="#neume072 #neume075"/>
  </neume>
  <formula type="with-gorgon" startid="#neume072" endid="#neume075"/>
</syllable>
```

Gorgon effect: previous neume and the first of the present one

Final cadence

```xml
<syllable>
  <syl>te-</syl>
  <neume xml:id="neume173" dur="4" dots="1" intmb="0" pname="d" slur="i">
    <neume xml:id="neume174" func="melodic" name="ison" pos="main"/>
    <neume xml:id="neume175" func="temporal" name="klasma" pos="above"/>
  </neume>
  <neume xml:id="neume176">
    <neume xml:id="neume177" func="melodic" name="iporroi" pos="main">
      <note xml:id="neume177-note001" dur="8" intmb="-1" intm="-2" pname="c" slur="m"/>
      <note xml:id="neume177-note002" dur="4" intmb="-1" intm="-2" pname="b" slur="t"/>
    </neume>
    <neume xml:id="neume178" name="gorgon" pos="above" effect="#neume173 #neume177-note001"/>
  </neume>
  <neume xml:id="neume179" func="melodic" name="apostrofos" dur="8" intmb="-1" intm="-2"
   pname="a" slur="i"/>
  <neume xml:id="neume180">
    <neume xml:id="neume181" name="dyo-kendimata" dur="8" intmb="+1" intm="+2" pname="b"
     pos="below" slur="m"/>
    <neume xml:id="neume182" name="oligon" dur="2" dots="1" intmb="+1" intm="+2" pname="c"
     pos="main" slur="t"/>
    <neume xml:id="neume183" name="triimiargon" pos="above" effect="#neume179 #neume181
     #neume182"/>
  </neume>
  <formula type="with-argon" startid="#neume173" endid="#neume180"/>
</syllable>
```